

ROVANIEMI APUKKA, Finland

WMO: 028130

Lat: 66.579N

Lon: 26.011E

Elev: 107

StdP: 100.04

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.6	(c) -29.2	(d) -35.9	(e) 0.1	(f) -32.3	(g) -32.3	(h) 0.2	(i) -29.0	(j) 7.2	(k) -2.5	(l) 6.3	(m) -4.2	(n) 0.5	(o) 290	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 25.5	(d) 16.9	(e) 23.3	(f) 16.0	(g) 21.4	(h) 15.2	(i) 18.6	(j) 23.2	(k) 17.4	(l) 21.7	(m) 16.1	(n) 19.7	(o) 2.6	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.3	20.9	15.6	11.2	18.9	14.6	10.5	17.7	53.3	23.4	49.3	22.0	45.6	19.8	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.4	5.6	5.0		-35.5	26.9	3.8	2.4	-38.2	28.6	-40.4	30.0	-42.5	31.4	-45.2	33.1
				-35.0	20.1	3.1	1.4	-37.3	21.1	-39.1	21.9	-40.8	22.7	-43.1	23.6
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-13.0	-11.3	-7.0	-0.3	6.8	11.6	15.1	12.5	8.1	0.9	-4.2	-8.6		
	DBStd	10.83	7.32	8.56	6.02	4.26	4.40	3.87	3.26	3.34	3.29	4.10	5.71	7.16		
	HDD10.0	3671	714	597	527	311	119	26	2	14	75	283	427	577		
	HDD18.3	6361	972	830	785	561	357	205	108	182	308	542	677	836		
	CDD10.0	364	0	0	0	0	21	74	160	92	17	0	0	0		
	CDD18.3	14	0	0	0	0	0	3	8	2	0	0	0	0		
	CDH23.3	187	0	0	0	0	11	35	117	24	0	0	0	0		
	CDH26.7	29	0	0	0	0	1	6	19	3	0	0	0	0		
(14) Wind	WSAvg	2.2	1.7	2.0	2.4	2.4	2.8	2.7	2.3	2.2	2.2	2.2	1.9	2.0		
(15) Precipitation	PrecAvg	535	37	30	30	27	33	51	61	70	60	54	45	36		
	PrecMax	723	93	73	73	85	90	115	133	139	123	108	87	58		
	PrecMin	394	9	11	2	6	5	8	12	18	7	25	13	8		
	PrecStd	84	19	17	17	17	17	29	32	30	28	23	16	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	4.3	6.2	15.3	24.7	27.1	29.0	26.8	18.4	10.4	5.9	2.8		
		MCWB	0.7	2.5	2.6	8.0	13.4	17.5	19.9	19.4	14.0	8.9	5.6	2.3		
	2%	DB	-0.1	1.8	4.1	10.2	21.8	23.5	26.5	22.9	16.6	9.0	4.6	1.5		
		MCWB	-0.3	0.9	1.1	4.9	13.5	15.8	17.6	16.2	12.7	7.5	4.1	1.1		
	5%	DB	-1.2	0.6	2.5	7.9	18.3	21.2	24.2	20.7	15.2	7.9	3.2	0.7		
		MCWB	-1.4	0.1	0.4	3.8	11.1	14.2	16.8	15.3	12.1	6.8	2.8	0.4		
	10%	DB	-2.7	-0.5	1.2	6.0	15.3	18.9	22.2	18.8	13.6	6.7	2.0	0.1		
		MCWB	-3.1	-0.9	-0.1	2.6	9.7	12.9	16.1	14.6	11.5	5.8	1.7	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	2.7	3.1	8.2	14.9	18.6	20.6	19.9	14.9	9.7	5.6	2.5		
		MCDB	1.3	3.9	5.7	14.4	21.9	24.0	25.2	25.5	17.1	10.0	5.7	3.0		
	2%	WB	-0.3	1.1	1.7	5.6	13.4	17.1	19.2	17.7	13.6	7.8	4.0	1.1		
		MCDB	-0.1	1.5	3.4	9.4	20.8	22.9	23.9	21.1	15.5	8.5	4.4	1.3		
	5%	WB	-1.5	0.2	0.8	4.1	12.0	15.6	18.1	16.2	12.7	6.8	2.8	0.5		
		MCDB	-1.3	0.6	1.9	7.3	17.2	19.3	22.2	18.9	14.5	7.7	3.1	0.7		
	10%	WB	-3.1	-0.9	0.1	3.0	10.2	13.9	16.9	15.2	11.7	5.9	1.7	-0.1		
		MCDB	-2.7	-0.5	1.1	5.4	14.3	17.2	21.2	17.7	13.2	6.7	2.0	0.1		
(35) Mean Daily Temperature Range	MDBR		8.6	9.1	11.6	9.8	9.9	9.1	9.8	9.3	7.7	5.5	5.5	7.5		
	5% DB	MCDBR	9.6	6.0	9.8	12.6	16.1	13.5	14.2	13.0	11.1	6.9	4.5	5.9		
		MCWBR	9.7	5.4	7.6	7.9	8.2	6.4	6.9	7.3	7.4	5.8	4.4	5.9		
	5% WB	MCDBR	9.6	5.6	6.7	11.2	14.5	11.1	12.1	10.3	8.4	6.2	4.5	5.7		
		MCWBR	9.8	5.2	5.5	7.3	7.8	5.9	6.5	6.4	6.0	5.4	4.5	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.197	0.265	0.298	0.332	0.344	0.345	0.359	0.342	0.317	0.289	0.347	N/A		
	taud		1.933	2.103	2.164	2.238	2.445	2.471	2.450	2.481	2.533	2.345	2.655	N/A		
	Ebn at Noon		199	569	749	822	858	864	837	816	755	559	213	0		
	Edh at Noon		15	53	85	103	94	94	93	82	61	43	16	0		
(44) All-Sky Solar Radiation	RadAvg		0.07	0.56	2.00	3.71	4.74	5.16	4.83	3.51	1.84	0.63	0.13	0.00		
	RadStd		0.01	0.10	0.20	0.33	0.42	0.38	0.47	0.32	0.19	0.12	0.02	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page